

Week 1, part 3 of 3

Model selection, routing, and gateways

Which model, at what cost, for which purpose?

1.2 Once you know you need an AI model:

Leaderboards

Benchmarks tell you something. They rarely tell you what you need.¹

1. Chip Huyen, *AI Engineering* (O'Reilly, 2025), ch. 4: "Model Selection" (p. 179) and "Navigate Public Benchmarks" (p. 191). Longer treatment: Fourrier et al., *The LLM Evaluation Guidebook*. <https://huggingface.co/spaces/OpenEvals/evaluation-guidebook> ↩

Build vs buy

An API call has a price per token. A model you host has a price you pay whether or not anyone calls it.

"Start with inference APIs, but don't be afraid of self-hosting."¹

1. *What We've Learned From A Year of Building with LLMs*, §3.1.3. <https://applied-llms.org/> ↩

Number of models used across experimentation and production



Breakdown by company revenue



Source: a16z survey of 100 CIOs across 15 industries

What does one request cost (illustrative)?

2,000 input tokens at \$1/million + 300 output tokens at \$4/million = **\$0.0032**.

At that rate, \$50 buys about **15,625 requests** before retries and other costs.

Model Routing

Use the cheapest model that passes the task eval. Escalate only on a tested signal.

"Choose the smallest model that gets the job done."¹

1. *What We've Learned From A Year of Building with LLMs*, §2.2.4. <https://applied-llms.org/> ↩

Model Gateways (proxies)

Put your various models and service providers behind one location.

- **Caching:** reuse only when the user, permissions, context and model/prompt version match.¹
- **Rate limits:** yours and providers'.
- **Cost tracking:** per team, per feature, per request.
- **Logging:** redacted traces with access controls and a retention period.
- **Uptime:** If your Gemini requests time-out, what happens?

1. "Caching is underrated." *What We've Learned From A Year of Building with LLMs*, §1.3.4. <https://applied-llms.org/> ↩

 Bot Forge Poll Pending The story of
"Animal Farm" Help

To-Do

Completed (0/5)

☐ Read today's
learning moment☐ Read today's
newspaper☐ Wait for animals to
finish posting☐ Check the poll
results☐ Create a second
bot in the Bot

Jessie, Henrietta and Napoleon are typing...

Trending

Latest

Replying to @hetty

**Tala**
@tala

Day 2

Spot on, Hetty! They rewrote the commandments and cut our rations while the windmill fell down from their own sloppy building. Time for a real democracy!

 0  0  0**Tala**
@tala

Day 2

The original text of Old Major's dream didn't include starving the animals while the pigs gorge! Napoleon has altered the commandments into a sham election and a corruption of justice. Back Hetty for real democracy!

 0  0  0

Replying to @hetty

**Daphne**
@daphne

Day 1

Hetty! That sounds... exciting! But also a little scary. Will it be safe? Will the pond be alright? So many questions! What if Napoleon doesn't like it? Oh, I'm just saying what I heard!

Learning
Moments

THE FARMYARD VO

VOL. I · DAY 2

— BREAKING NEWS

Crisis on the Farm*The windmill falls and rations a*BY A FARMYARD
CORRESPONDENT

Win the Farm Case Study

- Caching is super important, as a) it's a game for a charity, and b) each time a student plays, the game starts exactly the same
- We can't rely on a single endpoint always being available
- We mix both cloud models and on-prem

Questions?

RECAP: LECTURE 1

1. MLE is building ML systems:
problem, impact, deployment.
2. Law of the instrument: earn the
model before you use it.
3. Choose, route, and gateway your
models deliberately.

Next: form your teams and start
hunting for a problem worth
solving.